Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Steps to Solve	1. Distribute (if necessary) 2. Combine like terms. 3. Move variables to one side. 4. Solve!

Examples

1. $5y - 8 = 3y + 12$

$-3y \quad -3y$

$2y - 8 = 12$

$+8 \quad +8$

$\frac{2y}{2} = \frac{20}{2}$

$\boxed{y = 10}$

2. $-6x + 14 = 12 - 8x$

$+8x \quad +8x$

$2x + 14 = 12$

$-14 \quad -14$

$\frac{2x}{2} = \frac{-2}{2}$

$\boxed{x = -1}$

3. $7k = 3k - 36$

$-3k \quad -3k$

$\frac{4k}{4} = \frac{-36}{4}$

$\boxed{k = -9}$

4. $12 - 2u = 9u + 45$

$+2u \quad +2u$

$12 = 11u + 45$

$-45 \quad -45$

$\frac{-33}{11} = \frac{11u}{11}$

$\boxed{-3 = u}$

5. $11 - m = 51 - 6m$

$+6m \quad +6m$

$11 + 5m = 51$

$-11 \quad -11$

$\frac{5m}{5} = \frac{40}{5}$

$\boxed{m = 8}$

6. $-10k + 1 = 40 - 7k$

$+10k \quad +10k$

$1 = 40 + 3k$

$-40 \quad -40$

$\frac{-39}{3} = \frac{3k}{3}$

$\boxed{-13 = k}$

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7. $3(6p - 1) = 11p - 45$

$18p - 3 = 11p - 45$

$-11p \quad -11p$

$7p - 3 = -45$

$+3 \quad +3$

$\frac{7p}{7} = \frac{-42}{7}$

$\boxed{p = -6}$

8. $2(4w - 1) = -10(w - 3) + 4$

$8w - 2 = -10w + 30 + 4$

$8w - 2 = -10w + 34$

$+10w \quad +10w$

$18w - 2 = 34$

$+2 \quad +2$

$\frac{18w}{18} = \frac{36}{18}$

$\boxed{w = 2}$

9. $9(m - 3) + 3m = 7m + 43$

$9m - 27 + 3m = 7m + 43$

$12m - 27 = 7m + 43$

$-7m \quad -7m$

$5m - 27 = 43$

$+27 \quad +27$

$\frac{5m}{5} = \frac{70}{5}$

$\boxed{m = 14}$

10. $-16 - 7(2a + 3) = 23 - 2a$

$-16 - 14a - 21 = 23 - 2a$

$-14a - 37 = 23 - 2a$

$+14a \quad +14a$

$-37 = 23 + 12a$

$-23 \quad -23$

$\frac{-60}{12} = \frac{12a}{12}$

$\boxed{-5 = a}$

11. $5x - (x - 18) = 6 - 2(x + 15)$

$5x - x + 18 = 6 - 2x - 30$

$4x + 18 = -2x - 24$

$+2x \quad +2x$

$6x + 18 = -24$

$-18 \quad -18$

$\frac{6x}{6} = \frac{-42}{6}$

$\boxed{x = -7}$

12. $8(y + 4) - 2(y - 1) = 70 - 3y$

$8y + 32 - 2y + 2 = 70 - 3y$

$6y + 34 = 70 - 3y$

$+3y \quad +3y$

$9y + 34 = 70$

$-34 \quad -34$

$\frac{9y}{9} = \frac{36}{9}$

$\boxed{y = 4}$

Summary: _____

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